

A Four-Sector TESS Ephemeris for TOI-3505.01 and the Origin of a 2022 Ground-Based Null Observation

Mason Cao¹, Annalyse Dickinson², Owen Alfaro², Rianne Eccleston², Kasey Davidson², and Peter Plavchan²

¹Lambert High School, Suwanee, Georgia, USA

²Mason Space Center, Department of Physics and Astronomy, College of Science, George Mason University, 4400 University Drive, MS 3F3, Fairfax, VA 22030, USA

Context. Transit ephemerides lose predictive precision when their period or epoch is superseded, which can move a short ground-based observing sequence away from the event it was intended to cover. TOI-3505.01 (TIC 390988385) remains listed as a planet candidate.

Aims. We reanalyzed a 2022 George Mason University (GMU) 0.8 m R-band sequence, combined the four public TESS sectors, tested the stability of an updated linear ephemeris, and investigated why the ground sequence did not show the cataloged signal.

Methods. We performed differential aperture photometry on 263 quality-reviewed ground measurements and measured individual TESS events with a local linear baseline and an exposure-integrated trapezoid. The primary timing fit used one light curve per event: QLP for Sectors 14 and 41 and two-minute SPOC photometry for Sectors 54 and 81. We tested sector aggregation, event and sector deletion, selection thresholds, a quadratic timing model, and one public MuSCAT2 timing as an external control. We also propagated public 2021 timing markers with the superseded period recorded in a later ExoFOP report.

Results. The fixed 2022 schedule window gives a depth of -0.658 ± 0.395 ppt and therefore no transit-like dimming. The adopted ephemeris uses 27 TESS mid-transit times over 5.05 yr and gives $P = 2.9151516 \pm 0.0000049$ d and $T_0 = 2459793.531145653 \pm 0.0008800$ BJD_{TDB}. A quadratic term is not favored ($\Delta\text{BIC} = +1.71$). The current ephemeris places the relevant event 20.31 h before the 2022 schedule midpoint. In contrast, propagating the public 2021 ingress, midpoint, and egress by 96 cycles at the superseded 2.9174250 d period reproduces all three schedule markers within 56.1 s.

Conclusions. The 2022 null observation is consistent with a stale schedule window rather than evidence against the recurring TESS signal. The updated timing solution is robust to the tested controls, but the unresolved 0.517 arcsec companion and uncertain dilution prevent a physical radius measurement. This work does not validate or confirm TOI-3505.01.

Keywords: planetary systems · eclipses · ephemerides · photometry · methods: data analysis · stars: individual (TOI-3505)

Status boundary. The public snapshot used here lists both the TESS and TFOP Working Group dispositions as PC and PC, respectively. Throughout this paper, “candidate” is intentional; none of the analyses below constitutes statistical validation or confirmation.

1. Introduction

The Transiting Exoplanet Survey Satellite (TESS) searches bright stars for periodic decreases in flux produced when an orbiting body crosses the stellar disk (Ricker et al. 2015). TESS Objects of Interest (TOIs) are candidates that require additional scrutiny because an aperture-level transit signal can also be produced by stellar variability,

instrumental systematics, or an eclipsing source blended with the target. Ground-based time-series photometry can test whether an event occurs on the expected star, at the expected time, and with an approximately compatible depth and duration. Its interpretation depends on an ephemeris that is current enough to place the event inside the observing window.

TOI-3505.01 is associated with TIC 390988385 at $\alpha = 19:48:10.43$, $\delta = +18:41:56.1$ (ICRS; 297.043476° , 18.698914°). The TIC v8 snapshot gives $T = 10.9372$, an effective temperature of 6220 K, and a stellar radius estimate of $1.335 R_\odot$ (Stassun et al. 2019). The TOI catalog quantities frozen for this analysis are $P = 2.9151556 \pm 0.0000117$ d, epoch 2459793.534385 BJD_{TDB}, depth 2.91 ppt, and duration 2.004 h. A close companion reported in the project context is separated by 0.517 arcsec with $\Delta I = 1.7$ mag. That separation is far below a TESS pixel and below the effective resolution of the ground apertures used here.

The GMU program obtained an R-band sequence during the night of UT 2022 July 22. The recovered scheduling row placed ingress and egress inside that sequence, yet the reduced light curve did not show a transit. A null observation can have several explanations: inadequate precision, an astrophysical false positive, an event outside the data, or a stale prediction. The public record later acquired additional TESS sectors and ground reports, creating enough timing leverage to distinguish among those possibilities without treating any report-level result as an independent validation.

In this paper, we combine all four public TESS sectors available for TOI-3505.01, construct a 27-event timing solution, quantify its sensitivity to analysis choices, and reconstruct the historical 2022 schedule. To our knowledge, this is the first dedicated study to combine all four publicly available TESS sectors of TOI-3505.01 and trace the 2022 GMU null result to a schedule window consistent with a superseded ephemeris. This deliberately narrow novelty statement does not claim the first detection, classification, follow-up observation, validation, or confirmation of the candidate.

Section 2 inventories the target, TESS products, GMU sequence, and public follow-up context. Section 3 describes the ground and TESS analyses, timing controls, schedule reconstruction, and localization screens. Section 4 reports the measurements. Section 5 interprets the null observation and defines the limits of the candidate-level result. Section 6 summarizes the conclusions and future work.

2. Observations

2.1. Catalog and public status

We queried the TESS Input Catalog version 8 and Gaia Data Release 3 around the target (Stassun et al. 2019; Gaia Collaboration et al. 2023). The frozen TIC cone-search result contains a contamination ratio of 0.547. We use that number only as a warning that the aperture is blended; it is not a correction applied to a final transit depth. The Exoplanet Follow-up Observing Program for TESS (ExoFOP-TESS) target snapshot was retrieved on 2026 August 22. It lists no confirmed planet for this target and retains both relevant dispositions as planet candidate (PC).

2.2. TESS photometry

TOI-3505.01 has public TESS photometry in Sectors 14, 41, 54, and 81. We used MIT Quick-Look Pipeline (QLP) products in every sector for a like-for-like recovery and depth comparison (Huang et al. 2020a,b). For the primary timing analysis, QLP is the available series in Sectors 14 and 41; the higher-cadence SPOC PDCSAP light curves were selected in Sectors 54 and 81 (Jenkins et al. 2016). We retained finite measurements with finite errors and QUALITY = 0. QLP and SPOC reductions of a given sector are alternative measurements of the same TESS observations, not independent detections.

Table 1. TESS data used in the sector comparison and primary timing fit. Depths are the observed QLP aperture depths from the uniform first-pass box analysis and are not dilution corrected. The pipeline and cadence columns identify the series selected for the final event timing.

Sector	Public light-curve span (UTC)	Timing pipeline	Cadence (min)	Accepted timings	QLP depth (ppt)
14	2019 Jul 18–Aug 14	QLP	30	4	3.73 ± 0.42
41	2021 Jul 24–Aug 20	QLP	10	8	3.02 ± 0.21
54	2022 Jul 09–Aug 04	SPOC	2	9	2.68 ± 0.30
81	2024 Jul 15–Aug 10	SPOC	2	6	3.13 ± 0.24

The TESS products were obtained from the Mikulski Archive for Space Telescopes (MAST). The four quality-filtered QLP series span 2019 July to 2024 August, while the accepted event midpoints provide a 5.05 yr timing baseline. Official SPOC Data Validation reports are available for Sectors 54 and 81 and for a combined search using those same two sectors. We treat those products as same-observation method checks.

TOI-3505.01 in each TESS sector

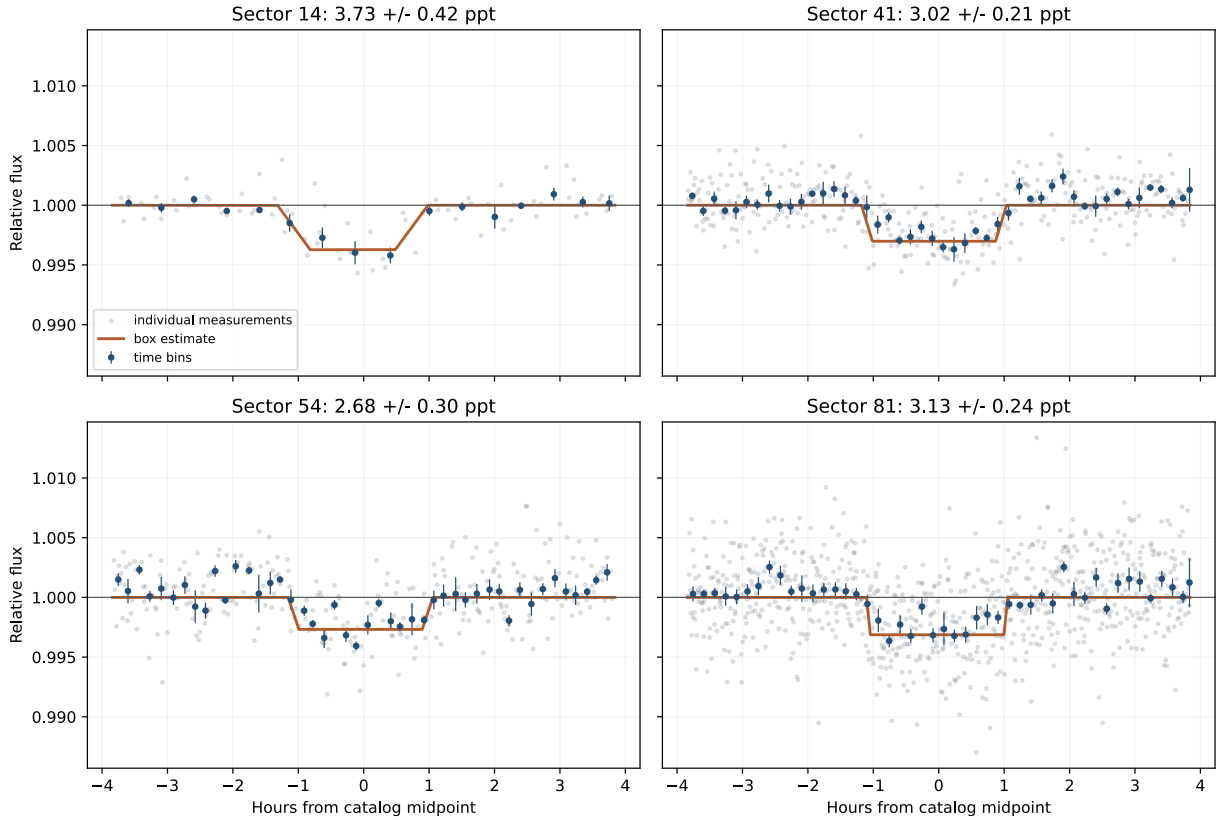


Figure 1. Uniform QLP recovery in all four TESS sectors, folded on the catalog ephemeris. Pale points are individual quality-filtered measurements, blue points are time bins, and the orange line is the exposure-integrated box estimate used for the first-pass sector depth comparison. The sector depths are observed aperture depths, not dilution-corrected physical depths. The final timing analysis replaces this box shape with an exposure-integrated trapezoid and uses SPOC two-minute photometry where available.

2.3. GMU 0.8 m sequence

The ground sequence was acquired through an R filter with 50 s exposures. The aligned analysis set contains 281 images from $\text{BJD}_{\text{TDB}} 2459782.598234811$ to 2459782.809458706 , a duration of 5.069 h. The schedule row gives local-clock ingress at 00:15 and egress at 01:54. The project owner confirmed that these clocks are Eastern time; America/New_York was EDT (UTC−4) on the observing night. The converted window is $\text{BJD}_{\text{TDB}} 2459782.682401\text{--}2459782.751151$ and is fully contained in the analyzed sequence.

The original workbook, its formulas and revision history, its Transit Info file, and a clock-synchronization record are not available. We preserve those missing records as provenance limitations rather than treating the schedule row as a complete ephemeris.

2.4. Public ground follow-up context

The ExoFOP snapshot contains seven time-series rows representing 6 unique observing nights; two rows duplicate the same GMU observation on UT 2021 June 28. The reports include earlier off-window observations, later reports describing on-target events, and a tentative MuSCAT2 egress. We did not re-reduce those external light curves. Only the MuSCAT2 report supplies a fitted midpoint uncertainty suitable for the named external timing control.

Table 2. Unique public ExoFOP time-series nights in the frozen 2026 August 22 snapshot. “Context” records report-level information; it does not mean that the underlying data independently validate the candidate.

UT date	Facility	Filter	Coverage	Use here
2021-06-28	GMU 0.8 m	R	Full	Context only
2021-10-15	ULMT 0.6 m	rp	Full	Context only
2022-06-14	KeplerCam 1.2 m	ip	Full	Report-level context
2023-05-05	CMO-SAI-RC-600 0.6 m	g-prime	Full	Report-level context
2023-07-11	OAA 0.4 m	lc	Full	Report-level context
2023-07-14	TCS 1.52 m	g, r, i, z_s	Egress	Timing control only

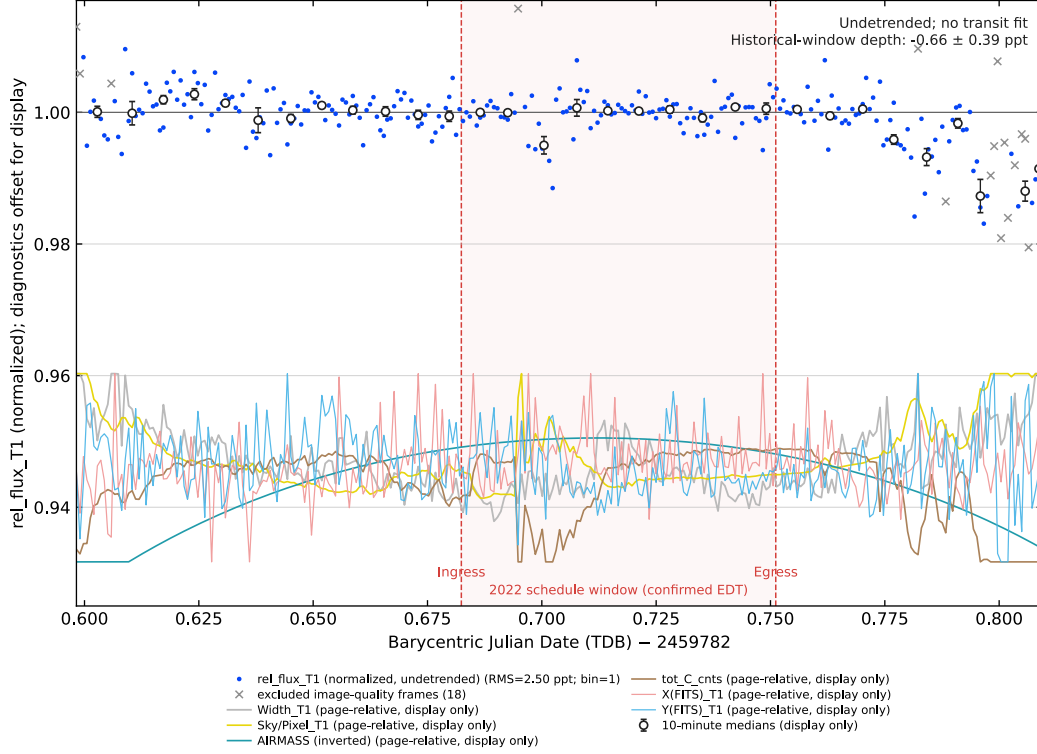
3. Analysis

3.1. Ground reduction and differential photometry

We reduced each science image as (science – median 50 s dark) / median normalized R flat. No separate bias subtraction, dark-time scaling, cosmic-ray filter, or interpolating registration was applied. We registered 281 frames to the first image using integer-pixel shifts derived from 25 reference stars. The median post-registration stellar-position RMS was 0.526 pixel.

We used AstroImageJ-style differential aperture photometry (Collins et al. 2017) with a 25 pixel source radius, a 70–139 pixel sky annulus, and the summed counts of ten fixed comparison stars, C2–C11. The final curve uses 263 measurements and excludes 18 measurements for predeclared condition failures or a field-wide image artifact. Target faintness by itself was never an exclusion criterion. The adopted relative flux was not detrended: none of the tested airmass, sky, width, position, comparison-count, centroid, or combined models passed the blocked cross-validation promotion rules.

We measured the ground precision with the robust scatter of the unbinned adopted flux. For the historical-window test, we fit a straight local baseline plus a fixed, exposure-integrated box between the schedule ingress and egress. The event time and duration were not optimized. We then injected a 2.91 ppt box at the same fixed times and repeated the fit. This same-window injection tests the arithmetic and the sensitivity of this light curve at the historical window; it is not a target-only pixel simulation and does not solve dilution.



Schedule times confirmed as Eastern; the current-catalog transit is outside this sequence.

Figure 2. Final GMU 0.8 m R-band differential light curve. Blue points are the 263 accepted measurements, open circles are ten-minute display medians, and gray crosses are excluded image-quality frames. The red interval is the confirmed-EDT 2022 schedule window. Width, sky, inverted airmass, comparison counts, and detector positions are scaled into the lower display band only; they do not enter the adopted flux or fixed-window depth. The legend is outside the data axes.

3.2. TESS event times and the linear ephemeris

The uniform sector analysis searched a local period and duration grid with a finite-exposure box and one straight baseline per event. For the final timing analysis, we used an exposure-integrated symmetric trapezoid. Its flat-to-total duration fraction, $T_{23}/T_{14} = 0.378$, was derived from the SPOC multi-sector radius ratio and impact parameter. This approximate shape represents the long ingress and egress of the near-grazing solution without claiming a limb-darkened physical fit.

Each event was measured within a ± 0.20 d local window. We accepted a midpoint if the local window was complete, its sampled-window coverage was at least 0.70, the fitted depth signal-to-noise ratio was at least 2.5, and its timing uncertainty was no greater than 0.03 d. These rules retained 27 midpoints: four in Sector 14, eight in Sector 41, nine in Sector 54, and six in Sector 81. QLP was used in Sectors 14 and 41, and SPOC two-minute data were used in Sectors 54 and 81, so each transit entered once.

We fit the accepted midpoints with

$$T(E) = T_0 + P E,$$

where E is the integer cycle relative to the catalog epoch. Weighted least squares used the event-level midpoint errors. When χ^2_{ν} exceeded one, we multiplied the covariance-derived parameter errors by $\sqrt{\chi^2_{\nu}}$; for the adopted fit

the scale is 1.141. We retained the epoch–period covariance when propagating prediction uncertainties. Times are reported in BJD_{TDB} ; conversions followed the barycentric timing conventions of Eastman et al. (2010) and used Astropy (Astropy Collaboration et al. 2022).

3.3. Timing robustness and model comparison

We evaluated whether the period depended on individual events, sector sampling, event-selection thresholds, or model complexity. First, we collapsed each sector to one conservative timing anchor; each anchor uncertainty was the larger of its formal weighted-mean error and its empirical within-sector standard error. Second, we refit after omitting each sector and after omitting each accepted event. Because there are only four sector clusters, the delete-one-sector jackknife is a sensitivity scale rather than a calibrated replacement for the primary uncertainty. Third, we repeated the fit at depth-S/N thresholds of 3 and 4.

We compared the linear ephemeris with a pivoted quadratic timing model,

$$T(E) = a + b(E - E_p) + q(E - E_p)^2,$$

using the Bayesian information criterion (BIC; Schwarz 1978). Finally, we appended the public MuSCAT2 midpoint, $\text{BJD}_{\text{TDB}} 2460140.4350453 \pm 0.000639$, to a separate 28-point control. The report describes a tentative egress, so this point never enters the headline TESS-only result.

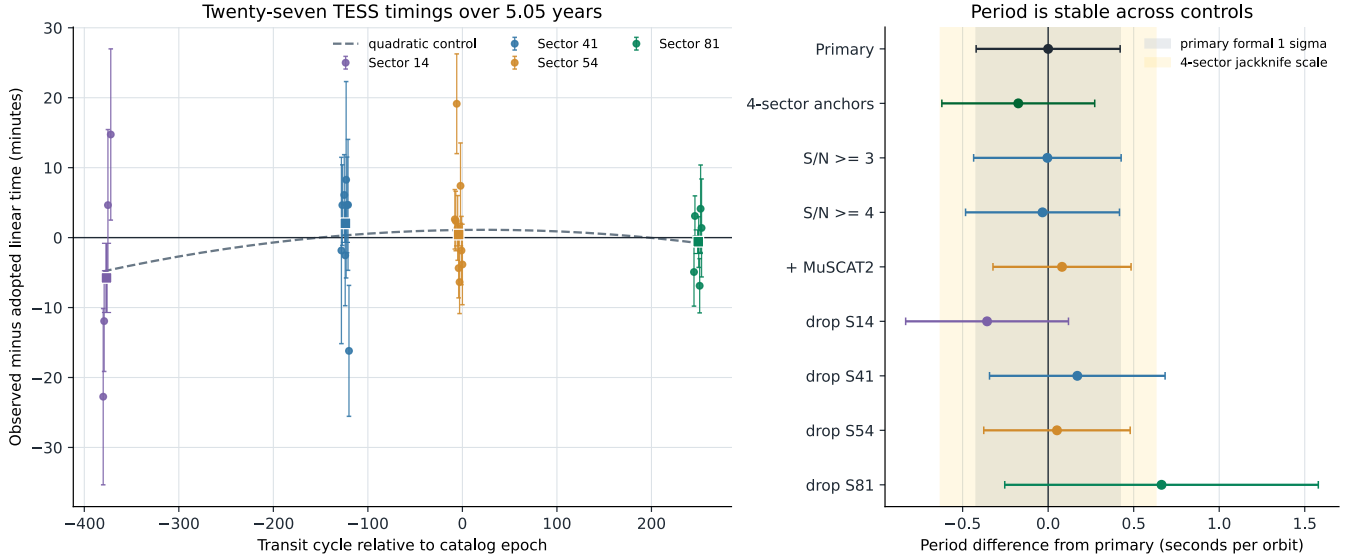


Figure 3. Ephemeris robustness. Left: observed minus adopted-linear timing residuals for 27 accepted TESS events over 5.05 yr; squares are conservative sector anchors and the dashed curve is the quadratic control. Right: period differences from the primary TESS-only fit. The gray band is the primary formal 1σ interval and the gold band is the four-sector jackknife sensitivity scale. QLP and SPOC entries are not independent when derived from the same sector.

3.4. Reconstruction of the 2022 schedule

The schedule row visibly lists 2.9151488 d but does not preserve its prediction epoch. We therefore searched the public report history for a prior event whose propagated ingress, midpoint, and egress could generate the schedule cells. A 2021 October 15 ULMT report gives all three BJD_{TDB} markers. A later CMO report records both a superseded period of 2.9174250 d and the revised 2.9151488 d period, and states that the revised Sector 41 ephemeris moved prior events by approximately 20 h.

We propagated each ULMT marker by the same integer number of cycles,

$$T_{2022, \text{marker}} = T_{2021, \text{marker}} + 96 \times 2.9174250 \text{ d},$$

and compared the result directly with the confirmed-EDT schedule markers. As controls, we propagated the same 2021 midpoint with the revised period and calculated the nearest event from the adopted TESS ephemeris. This is an archival consistency reconstruction. Without the original workbook, agreement cannot distinguish a stale formula from manually copied timing cells.

Table 3. Marker-by-marker schedule reconstruction. Positive residuals mean the schedule marker is later than the propagated marker.

Marker	2021 BJD _{TDB}	+96 cycles	2022 schedule	Schedule – propagation (s)
Ingress	2459502.6093000	2459782.6821000	2459782.6824006	+26.0
Midpoint	2459502.6441000	2459782.7169000	2459782.7167758	-10.7
Egress	2459502.6790000	2459782.7518000	2459782.7511510	-56.1

TOI-3505.01: archival reconstruction of the 2022 schedule

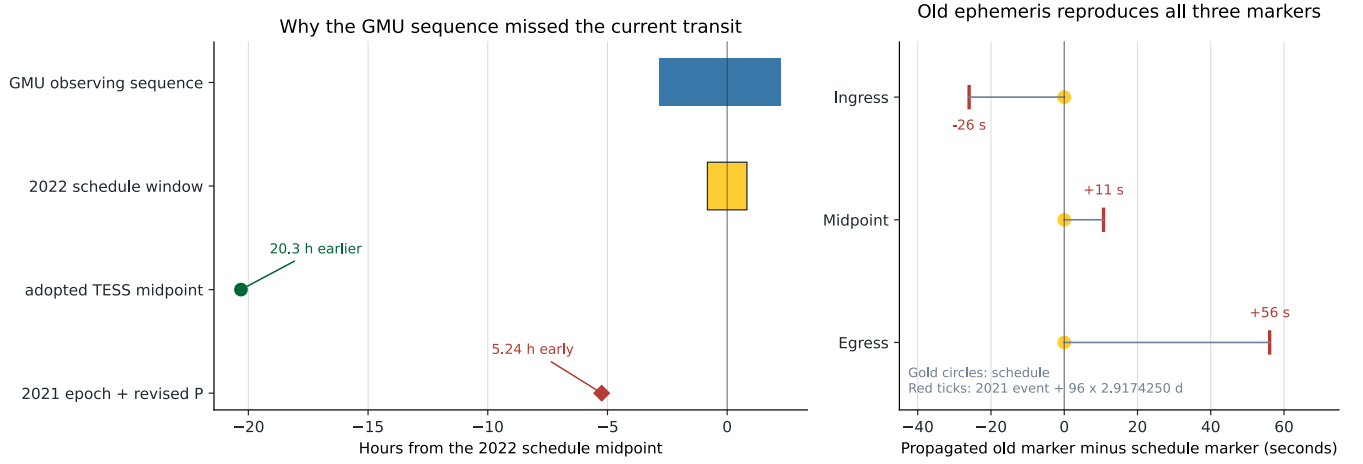


Figure 4. Archival reconstruction of the 2022 schedule. Left: the GMU sequence covered the schedule window, while the adopted TESS event occurred 20.31 h before the schedule midpoint; applying the visible revised period to the 2021 midpoint predicts a time 5.24 h early. Right: propagating each 2021 marker by 96 cycles at 2.9174250 d reproduces ingress, midpoint, and egress within 56.1 s. The match is strong evidence of consistency, not proof of the missing workbook history.

3.5. Localization and contamination screens

We compared circular TESScut apertures from 1.5 to 4.0 pixels and constructed difference images at a fixed 3.0 pixel reference radius. In every sector, the difference centroid lies within one TESS pixel of the target coordinates. This localizes the signal to the target system at TESS resolution but cannot distinguish the primary star from the 0.517 arcsec companion.

For the ground images, we measured all 280 TIC sources within 2.5 arcmin that were bright enough to mimic a 2.91 ppt event under a total-eclipse screen. At the confirmed schedule window, 222 sources were inconsistent with the required dimming at the screen’s three-sigma criterion, while 2 source apertures overlap the target aperture. This is a conditional project screen, not the formal program AstroImageJ nearby-eclipsing-binary procedure. Neither it nor the TESS difference images resolve the close companion.

We also calculated dilution scenarios using the TIC contamination ratio and ΔI as a rough TESS-band proxy. Because the QLP crowding treatment and the companion’s actual TESS-band contrast are not resolved, we did not apply those scenarios as a final correction and do not report a planet radius.

4. Results

4.1. Ground light curve

The 263 accepted ground measurements have an unbinned robust scatter of 2.497 ppt. The fixed historical-window fit gives -0.658 ± 0.395 ppt, or -1.67σ when positive depth denotes dimming. The negative value is therefore not a transit-like event. The adopted current-catalog prediction was already 17.39 h before the first accepted ground exposure, so the ground sequence did not observe the current predicted event.

Adding a 2.91 ppt same-window box recovered an incremental depth of 2.910 ppt. Including the measured negative control at that location, the total recovered dimming has signal-to-noise 5.71. The injection demonstrates sensitivity to a catalog-scale dimming at the fixed historical window; it does not establish sensitivity to every transit shape or blend scenario.

4.2. Four-sector signal and adopted ephemeris

The uniform QLP analysis recovers a recurring dip in all four sectors with observed depths from 2.68 to 3.73 ppt (Figure 1). These values are consistent at the level expected for different cadences and apertures, but they are not physical, dilution-corrected depths. The official SPOC reports for Sectors 54 and 81 recover the same signal and do not show a strong odd/even, secondary-event, or centroid alert. Because they reuse the same TESS pixels, those reports are diagnostic comparisons rather than independent evidence.

The adopted linear fit is

$$T_0 = 2459793.531145653 \pm 0.0008800 \text{ BJD}_{\text{TDB}}, \quad P = 2.9151515671 \pm 0.0000048720 \text{ d}.$$

It has $\chi^2_{\nu} = 1.301$ after the stated uncertainty scaling and an RMS timing residual of 8.51 min. Its formal period uncertainty is 2.40 times smaller than the frozen TOI catalog uncertainty and 1.43 times smaller than the SPOC multi-sector value. Precision comparisons do not make the fit independent of those source observations.

Table 4. Linear ephemeris estimates. The primary row is the only headline result. The four-sector and MuSCAT2 rows are controls with the limitations described in Section 3.

Fit	N	Period (d)	1 σ error (d)	χ^2_{ν}	Role
Primary TESS linear	27	2.9151515671	0.0000048720	1.30	Adopted
Four sector anchors	4	2.9151495461	0.0000051714	0.49	Conservative aggregation
QLP only	24	2.9151569718	0.0000062841	1.10	Pipeline control
TESS + MuSCAT2	28	2.9151525056	0.0000046714	1.28	External control only

4.3. Robustness controls

The four conservative sector anchors give $P = 2.9151495461 \pm 0.0000051714$ d, consistent with the primary result. The delete-one-sector jackknife scale is 0.0000072894 d, or 1.50 times the primary formal error. We do not substitute this four-cluster value for the event-level uncertainty. The delete-one-event jackknife scale is 0.0000057767 d, and no single-event deletion shifts the period by more than 0.0000031595 d. Raising the minimum depth S/N to 3 or 4 leaves the period within the primary formal uncertainty.

The quadratic coefficient is $q = -2.544\text{e-}08 \pm 2.295\text{e-}08$ d cycle $^{-2}$ (-1.11σ). It improves χ^2 by only 1.58 while adding a parameter, yielding $\Delta\text{BIC} = \text{BIC}_{\text{quadratic}} - \text{BIC}_{\text{linear}} = +1.71$. The timing data therefore do not favor curvature. Adding the tentative MuSCAT2 timing only as an external control gives $2.9151525056 \pm 0.0000046714$ d, a 4.12% formal precision change, and remains consistent with the TESS-only period.

4.4. Historical schedule

The adopted ephemeris places the relevant midpoint 20.31 h before the schedule midpoint. Propagating the public 2021 markers by 96 cycles with the superseded 2.9174250 d period instead reproduces the 2022 schedule: the

ingress, midpoint, and egress residuals have maximum absolute magnitude 56.1 s (Table 3). Applying the visible 2.9151488 d value to the same 2021 midpoint predicts an event 5.24 h before the schedule midpoint. The combined evidence is strongly consistent with timing cells retained from the superseded ephemeris while the displayed period cell was updated.

5. Discussion

5.1. Interpretation of the 2022 null

The ground light curve alone answers a limited question: no transit-like dimming is measured inside the exact window the 2022 schedule told the observers to cover. The same-window injection shows that a catalog-scale box would have been recovered in that portion of the data. The updated TESS ephemeris then resolves the apparent conflict by placing the actual predicted event outside the sequence.

The schedule reconstruction provides a specific historical explanation. The three marker cells are jointly reproduced to better than one minute by a public 2021 prediction propagated with the superseded period. A coincidental match of ingress, midpoint, and egress is less plausible than a single midpoint coincidence because it also preserves the event duration. Nevertheless, we cannot inspect the missing workbook formulas or revision history. We therefore describe the result as a strongly supported archival reconstruction, not proof that a particular spreadsheet formula caused the error.

The null observation should not be used as evidence that the TESS signal disappeared. It is an off-transit observation under the adopted ephemeris. Conversely, explaining the schedule does not turn the ground data into a transit detection.

5.2. Timing precision and model scope

The long, four-sector baseline supplies most of the period leverage. Sectors 14 and 81 are separated by more than five years, while the two-minute SPOC sampling in the later sectors improves individual midpoint precision. The stability of the period under conservative sector aggregation, stricter selection, event deletion, sector deletion, QLP-only processing, and the external MuSCAT2 control argues that the headline period is not an artifact of one event or one threshold.

The formal uncertainty remains conditional on the approximate trapezoid, the event-level timing errors, and the single linear ephemeris. The four-sector jackknife is larger but is based on only four clusters. Reporting both quantities distinguishes a fit uncertainty from a coarse sensitivity scale. The positive ΔBIC for the quadratic model means that these timings do not justify an added curvature parameter; it does not prove that the physical period is exactly constant.

5.3. Candidate status and unresolved blends

The recurring signal, weak odd/even and secondary-event diagnostics, and TESS-scale centroid agreement are all compatible with a transit on the target system. They do not determine which component of the unresolved system is dimming. A 0.517 arcsec companion is inseparable in the TESS images and in the 25 pixel ground aperture. The conditional nearby-star screen also leaves 2 apertures overlapping the target and is not a formal nearby-eclipsing binary clearance.

These limitations are why we do not apply the scenario dilution correction, infer a planet-to-star radius ratio, calculate a planet radius, or report a false-positive probability. The live public status at the analysis date remains PC. Statistical validation would require a documented combination of resolved imaging, spectroscopy or radial velocities as appropriate, fully characterized light curves, and a formal false-positive analysis beyond the scope of this work.

5.4. Literature and novelty boundary

TOI-3505.01 appears in the population-level Ariel target list of Edwards & Tinetti (2022), and public ground follow-up predates this analysis. Exact searches for “TOI-3505.01,” “TOI 3505,” and TIC 390988385 found no dedicated paper or preprint in the dated literature snapshot. The supported novelty is therefore the combination of all four public TESS sectors with the archival schedule reconstruction, not discovery or first follow-up. Because a search can miss an unexpected identifier or unindexed venue, the phrase “To our knowledge” must remain, and the search should be repeated immediately before submission or preprint posting.

6. Conclusions and future work

We analyzed the GMU 0.8 m R-band sequence and all four public TESS sectors of TOI-3505.01. The ground data show no dimming in the confirmed-EDT 2022 schedule window: -0.658 ± 0.395 ppt. The primary TESS-only fit gives $P = 2.9151516 \pm 0.0000049$ d from 27 events over 5.05 yr, and the tested resampling, selection, pipeline, external-timing, and quadratic controls do not motivate a different headline solution. The adopted event was 20.31 h before the schedule midpoint, while a superseded 2.9174250 d public ephemeris reproduces all three schedule markers within 56.1 s. The simplest supported interpretation is that the 2022 observations followed a stale event window.

Future work should obtain a new, full transit using the updated prediction and an independently verified observatory clock; repeat the formal nearby-eclipsing-binary procedure; measure the close companion’s contrast in or near the TESS band; and acquire resolving imaging and spectroscopic constraints needed for a formal false-positive assessment. A joint, exposure-integrated, limb-darkened transit model should be attempted only after the dilution model is documented. Recovering the original 2022 workbook, formula history, Transit Info file, and clock record would further strengthen the archival interpretation. Until those steps are complete, TOI-3505.01 remains a planet candidate and no dilution-corrected radius should be quoted.

Acknowledgments

This work was completed through the Schar Scholars NASA Data Science Analysis and Scholarly Research Paper Writing internship at George Mason University. This paper includes data collected with the TESS mission and obtained from MAST at the Space Telescope Science Institute. Funding for U.S. institutions participating in TESS is provided by the NASA Explorer Program. STScI is operated by the Association of Universities for Research in Astronomy, Inc., under NASA contract NAS 5-26555. We acknowledge the use of TESS High-Level Science Products produced by the MIT Quick-Look Pipeline and made available by MAST. This research has made use of the ExoFOP-TESS website, which is operated by the NASA Exoplanet Science Institute at Caltech/IPAC under contract with NASA, and of data from the European Space Agency Gaia mission processed by the Gaia Data Processing and Analysis Consortium.

Data and code availability

The analysis code, frozen catalog and public-report snapshots, derived tables, figures, claim-evidence ledger, decision log, software record, and SHA-256 file manifest are available in the project repository at <https://github.com/mason-cao/nasa-tess-exoplanet-analysis>. TESS mission products were obtained from <https://archive.stsci.edu/missions-and-data/tess>; QLP products are identified by dataset DOI 10.17909/t9-ro86-e880. The public follow-up source is <https://exofop.ipac.caltech.edu/tess/target.php?id=390988385>. The repository preserves exact product filenames and retrieval metadata. The external reports are contextual; their image-level data were not re-reduced in this work.

Software

The reproducible workflow uses Python, NumPy, pandas, SciPy, Matplotlib, Astropy, astroquery, photutils, Lightcurve, and AstroImageJ. Exact installed versions and the source revision are recorded in `outputs/toi3505_research_record/software_versions.json`. All displayed manuscript values are injected from canonical machine-readable outputs by `src/build_toi3505_paper.py`; the paper build also writes an auditable `manuscript_values.json` record.

References

- Astropy Collaboration et al. 2022, “The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package,” *ApJ*, 935, 167, doi:10.3847/1538-4357/ac7c74.
- Collins, K. A., Kielkopf, J. F., Stassun, K. G., & Hessman, F. V. 2017, “AstroImageJ: Image Processing and Photometric Extraction for Ultra-precise Astronomical Light Curves,” *AJ*, 153, 77, doi:10.3847/1538-3881/153/2/77.
- Eastman, J., Siverd, R., & Gaudi, B. S. 2010, “Achieving Better Than 1 Minute Accuracy in the Heliocentric and Barycentric Julian Dates,” *PASP*, 122, 935, doi:10.1086/655938.
- Edwards, B., & Tinetti, G. 2022, “The Ariel Target List: The Impact of TESS and the Potential for Characterizing Multiple Planets within a System,” *AJ*, 164, 15, doi:10.3847/1538-3881/ac6bf9.
- Gaia Collaboration et al. 2023, “Gaia Data Release 3: Summary of the Content and Survey Properties,” *A&A*, 674, A1, doi:10.1051/0004-6361/202243940.
- Huang, C. X., Vanderburg, A., Pál, A., et al. 2020a, “Photometry of 10 Million Stars from the First Two Years of TESS Full Frame Images: Part I,” *RNAAS*, 4, 204, doi:10.3847/2515-5172/abca2e.
- Huang, C. X., Vanderburg, A., Pál, A., et al. 2020b, “Photometry of 10 Million Stars from the First Two Years of TESS Full Frame Images: Part II,” *RNAAS*, 4, 206, doi:10.3847/2515-5172/abca2d.
- Jenkins, J. M., Twicken, J. D., McCauliff, S., et al. 2016, “The TESS Science Processing Operations Center,” *Proc. SPIE*, 9913, 99133E, doi:10.1117/12.2233418.
- Ricker, G. R., Winn, J. N., Vanderspek, R., et al. 2015, “Transiting Exoplanet Survey Satellite,” *JATIS*, 1, 014003, doi:10.1117/1.JATIS.1.1.014003.
- Schwarz, G. 1978, “Estimating the Dimension of a Model,” *Annals of Statistics*, 6, 461, doi:10.1214/aos/1176344136.
- Stassun, K. G., Oelkers, R. J., Paegert, M., et al. 2019, “The Revised TESS Input Catalog and Candidate Target List,” *AJ*, 158, 138, doi:10.3847/1538-3881/ab3467.